



July 17, 2025

Problem 1

A drone must deliver packages to **8 locations** (A - H) in a city represented as a 2D grid. The drone **starts and ends at the Warehouse (W)**, and each location must be visited **exactly once**. Your task is to use a Genetic Algorithm (GA) to find an optimal route that minimizes the total distance traveled. The coordinates are provided below:

Location	X	Y
W (Warehouse)	0	0
A	2	3
B	5	4
C	3	7
D	6	1
E	8	6
F	1	8
G	7	3
H	4	2

Table 1: Coordinates of Delivery Locations

Tasks:

1. Propose a chromosome representation for this problem. Provide 4 example solutions (initial population) in the form of permutations.

[5 Marks]

2. Design a fitness function that minimizes the total route distance. Use the Euclidean distance formula between locations:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Hint: Since GAs maximize fitness, propose an appropriate transformation.

3. Apply **truncation selection** to retain the top 75% of the population. **Justify your choice.**

[5 Marks]

4. Perform **single point crossover** between:

- Individual 1 and Individual 3 (**fitness wise ascending**)
- Use crossover points at **positions 2**
- Generate and show **2 offspring** with mapping steps clearly

[5 Marks]

5. From the two offspring produced above:

- Select the **best one**
- Perform single point crossover with the **remaining selected individual**
- Use crossover points at **positions 3**
- Clearly show the final offspring after crossover

[5 Marks]

6. Select the best offspring from the previous step. Perform a mutation by **inverting the sub-sequence between position 3 and 6 (inclusive)**. Show the resulting mutated chromosome. [5 Marks]

Note:

- The route always starts and ends at the warehouse (W), which is implicit and should be included when calculating the total distance.

Theoretical Question (No Simulation Required):

In the previous question's crossover part, if we want to focus on **exploration**, what should we do differently compared to the previous crossover strategy? Briefly justify your answer based on Genetic Algorithm principles. [5 Marks]

Question 02

A company runs a product recommendation system for its e-commerce platform. Initially, recommendations are generated based only on customers' past purchase history. While this approach results in reasonable sales, the company notices that customer engagement growth has slowed and many customers feel the suggestions have become repetitive and less relevant.

The development team is now exploring ways to improve the recommendation strategy to increase both customer satisfaction and sales.

Questions

1. Define what is meant by a **local maximum** in the context of optimization problems. [2 Marks]
2. Explain how relying only on past purchase history for recommendations might cause the system to get stuck at a local maximum. [3 Marks]
3. The company decides to try a strategy where they periodically reset or vary the recommendation criteria to find better solutions. Explain how this approach can help overcome the stagnation problem and improve recommendations. [5 Marks]

